

West syndrome in an infant with vitamin B12 deficiency in the absence of macrocytic anaemia.

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Vitamin B(12) deficiency in infants often produces haematological and neurological deficits, including macrocytic anaemia, neurodevelopmental delay or regression, irritability, weakness, hypotonia, ataxia, apathy, tremor, and seizures. The diagnosis of vitamin B(12) deficiency can be difficult when the typical macrocytic anaemia is absent. We report the case of a 10-month-old female diagnosed with West syndrome associated with vitamin B(12) deficiency but without macrocytic anaemia caused by nutritional inadequacy in the mother. The patient's motor skills and cognitive development were normal until she was 9 months old, when she began to exhibit a series of sudden flexions of the head, trunk, arms, and legs. She was exclusively breast-fed and had received no vitamin supplementation. Results of electroencephalography (EEG) indicated modified hypsarrhythmia and the patient was diagnosed as having West syndrome. Synthetic adrenocorticotrophic hormone was administered and although her spasms had resolved, the patient remained apathic and could not sit without assistance. EEG results indicated generalized slow activity. After she was diagnosed as having vitamin B(12) deficiency, parenteral treatment with vitamin B(12) was initiated. Her symptoms resolved and EEG was completely normal. When she was 20 months old she exhibited an age-appropriate developmental and neurological profile. To our knowledge, this is the first report of West syndrome as a presenting symptom of vitamin B(12) deficiency.